ANGLO-CHINESE JUNIOR COLLEGE

JC2 Preliminary Examination

Higher 2

GEOGRAPHY **9173/01**

Paper 1 **28 August 2025**

3 hours

Marking Scheme
Suggested Answers

© ACJC 2025

[Turn over

Section A

Answer **both** questions.

Cluster 1: Development, Economy and Environment

Cluster 1: Development, Economy and Environment

1 Resource 1 shows the global oil production and consumption. Resource 2 shows an oil rig. Resource 3 shows historical oil prices relative to Organisation of the Petroleum Exporting Countries (OPEC) nations' Gross Domestic Product (GDP). Resource 4 shows the Human Development Index formula.

(a) Describe the pattern of global oil production and consumption as shown in Resource 1. [5]

Award 1 mark for each description of production/consumption pattern, to a maximum of 5 marks.
Award a maximum of one additional mark for further development of each description, where applicable.
Possible responses include:

Overall, oil production and consumption are unevenly distributed globally.
Production is concentrated in the Middle East, Russia, and North America, while consumption is highest in North America, Europe, and Asia [1 mark].
North America has very high oil consumption, especially the United States (~863 million oil-equivalent tons in 2016), making it the largest consumer worldwide [1 mark]. The US is also a leading producer, though it consumes more than it produces, creating import dependence [1 additional mark]. Canada, by contrast, produces over 200 million tons, more than it consumes, making it a net exporter [1 mark].
Middle East countries are major producers and exporters. For example, Saudi Arabia produced over 500 million tons but consumed less than 200 million tons [1 mark].
Asia is a key demand centre. China consumed 600 million tons, but its production was far lower, highlighting reliance on imports [1 mark].
Africa produced notable amounts of oil, especially in Nigeria, Libya, and Angola [1 mark]. However, much of Africa's production is exported, as domestic consumption remains low relative to output [1 additional mark].

(b) Using Resource 2, explain the impacts of extractive industries on the host economies. [6]

Award 1 mark for each valid explanation, to a maximum of 6 marks.
Award 1 additional mark for further development of each point, where applicable.

Possible responses include:

Environmental impacts (negative):

Extractive activities such as oil flaring release greenhouse gases (CO_2, methane) [1 mark]. This contributes to global warming and local air pollution, harming human health [1 additional mark].

Oil extraction can lead to land degradation and contamination of soil and water resources [1 mark]. This reduces agricultural productivity and threatens biodiversity [1 additional mark].

Positive economic impacts:

Oil extraction generates significant revenue through exports [1 mark]. This boosts government budgets and foreign reserves, funding welfare and development [1 additional mark].

Extractive industries create jobs both directly and indirectly. [1 mark]. This provides income opportunities and stimulates local economies [1 additional mark].

Infrastructure improvements:

Revenue from extractive industries is often invested in infrastructure such as roads, electricity, and pipelines [1 mark]. These developments can support wider economic diversification and social improvements [1 additional mark].

(c) Using Resource 3, describe the relationship between historical oil prices and the OPEC nations' GDP. [4]

Possible responses include:

There is a positive relationship between global oil prices and OPEC nations' GDP [1 mark]. When oil prices rose to 120 between 2011–2013, Saudi Arabia's GDP exceeded 700 billion [1 additional mark].

When oil prices fell, OPEC GDPs declined [1 mark]. For example, during the 2014-2015 drop in oil prices where prices fell to $43/barrel, Iran's GDP decreased by 200 billion. [1 additional mark].

Anomaly: Venezuela's GDP rose steadily from 2011-2015 even when oil prices were relatively decreased [1 mark]. This is seen where it went from 320 billion to 490 billion while oil decreased from $112/barrel to $90/barrel. [1 additional mark].

(d) Explain how countries can manage the impacts of extractive industries. [6]

Award 1 mark for each valid strategy, to a maximum of 6 marks.
Award 1 additional mark for further development with example or explanation.

Possible responses include:

Diversification of the economy reduces reliance on oil [1 mark]. For example, the UAE has invested in finance, trade, and tourism [1 additional mark].

Establishing sovereign wealth funds stabilises revenue across booms and busts [1 mark]. Norway's Government Pension Fund is a successful case [1 additional mark].

Environmental regulations reduce ecological impacts [1 mark]. For example, limiting gas flaring can lower carbon emissions [1 additional mark].

(e) Using Resource 4, explain the issues in measuring sustainable development. [4]

Award 1 mark for each valid issue, to a maximum of 4 marks.
Award 1 additional mark for further development with explanation.

Possible responses include:

HDI only captures health, education, and income [1 mark]. It ignores environmental sustainability and resource use [1 additional mark].

National averages mask inequalities within countries [1 mark]. For instance, urban–rural divides or gender gaps are hidden [1 additional mark].

Data reliability is inconsistent across nations [1 mark]. Poorer countries may lack accurate records [1 additional mark].

HDI excludes social or political dimensions like freedom and security [1 mark]. These are vital for understanding long-term sustainability [1 additional mark].

(f) Explain how variations over time and space of a provisioning service are influenced by demand and technology. [5]

Award 1 mark for each valid description, to a maximum of 5 marks.
Award a maximum of one additional mark for further development with example.
Award a maximum of 3 marks if only demand or only technology is addressed.

Possible responses include:

Demand: As populations grow, the demand for provisioning services such as food, fresh water, and timber increases [1 mark]. This often leads to agricultural expansion into new areas or more intensive use of existing land [1 additional mark].

Changing diets and lifestyles alter demand for services. For example, rising meat consumption in Asia has increased pressure on land and water for livestock and feed crops [1 mark].

Urbanisation increases demand for water and energy in cities, reshaping resource flows spatially [1 mark].

Technology determines access and volume of resources [1 mark]. For example, shale gas extraction in the USA since 2000 increased available energy and reduced reliance on imports [1 additional mark].

Resource classification shifts with technology: resources move from “potential” to “actual” when extraction becomes economically viable [1 mark]. For example, Canadian oil sands became profitable due to technological advances [1 additional mark].

Cluster 2: Tropical Environments

2 Resource 5 shows the sea surface temperature anomaly during an El Nino event in 2023. Resource 6 shows the spatial and temporal distribution of tropical cyclones. Resource 7 shows a meandering stream in the humid tropics. Resource 8 shows a braided stream located on a glacial outwash plain.

(a) With the aid of Resource 5, describe the oceanic and atmospheric conditions associated with an El Nino event. [4]

Award up to 2 marks for the oceanic conditions.
Award up to 2 marks for the atmospheric conditions.

Possible response:

Oceanic Conditions

- Warmer Ocean Temperatures:
  Surface waters in the central and eastern equatorial Pacific become significantly warmer than average. R5: +1.5^{o}C to 5°C
- Reduced Upwelling:
  The weakening of trade winds reduces the upwelling of cool, nutrient-rich waters from the deep Pacific Ocean along the west coasts of South and Central America.
- Warm Water Shift:
  Warm waters from the western Pacific, part of the "Warm Pool," move eastward toward the west coast of Equatorial South America.

Atmospheric Conditions

- Weakened Trade Winds:
- The prevailing easterly trade winds, which normally blow from east to west, weaken or can even reverse to a westerly direction
- Altered Atmospheric Circulation:
  The shift in ocean heat and pressure changes the balance of atmospheric circulation, altering the typical Walker Circulation. Low pressure is at the eastern Pacific ocean, while the high pressure location changes to the western Pacific.
- Increased Rainfall in the eastern Pacific: The warmer ocean water leads to increased convection and rainfall in the central and eastern Pacific.
- Drought in the West: Conversely, areas in the western Pacific, such as Indonesia, India, and Australia, experience reduced rainfall and conditions conducive to drought and wildfires.

Examiner's comments

- Need to include oceanic and atmospheric conditions and not only describe the sea surface temperatures (SSTs) of R5.
- ENSO events related to changes of pressure and rainfall is largely focused over the equatorial belt; so just need to focus all the description of oceanic and atmospheric conditions around the equatorial Pacific.
- As this is the description question, do not need to say what cause the observed patterns.
- It is also not accurate to say create the causal relationship between atmospheric and oceanic conditions. No scientist can attest to which is the original starting point of ENSO, be it atmospheric or oceanic. So do not say that due to Trade

wind reversals, it lead to high pressure on the eastern Pacific. Instead simply state each the conditions, but do not attribute them to a specific condition.

- When citing East or West, it must be relative to something. For example, it would the 'eastern Pacific Ocean' or off the west coast of South America.

(b) Explain the conditions leading to the spatial and temporal distribution of tropical cyclones shown in Resource 6. [6]

Award 1 mark for each condition provided
Award up to 2 marks for elaboration of the processes leading to tropical cyclone formation.

Possible response:

Conditions leading to spatial and temporal distribution:

1. Sea surface temperature (SST):
   Tropical cyclones form mainly between 5° and 25° N/S of the equator where sea temperatures exceed 26–27 °C, providing the heat and moisture needed.

   Seasonality of SST warming:
   Cyclones occur mainly in the late summer and early autumn of each hemisphere, when ocean waters have warmed to their maximum.
   Northern Hemisphere: June–November (peaks Aug–Sept).
   Southern Hemisphere: November–April (peaks Jan–Mar).

2. Coriolis force:
   They do not form within about 5° of the Equator because the Coriolis force is too weak to generate the necessary rotation. Hence, they are absent directly at the equator but common just outside it.
   Rotation generated by Coriolis force creates centrifugal forces which, in turn, creates an intense low pressure.
   The intense low pressure then continues to draw warm moist air into the tropical cyclone supplying the latent heat that keeps the air buoyant.

3. Triggers incipient storms (initial atmospheric disturbance) created by Trade winds convergence and the tropical easterly wave: These start off the formation of tropical cyclones by inducing the air to rise. Tropical Easterly waves tend to form off the tropical west coast of Africa, accounting for the formation of tropical cyclones over the Atlantic Ocean (spatial distribution). Over the Atlantic, about 60% of tropical cyclones and 85% of intense hurricanes originate from tropical easterly waves.

4. High humidity in the mid-atmosphere: The atmosphere needs to be unstable, which means that rising parcels of air continue to rise, allowing for deep convection.

5. Low vertical wind shear: Low wind shear allows the storm's vertical structure to remain intact, preventing it from being torn apart, allowing for the chimney effect which, in turn, allows for continued latent heat release and rising air.

Also to mention that the tropical cyclone tracks are the result of the easterly Trade winds arising from the convergence of the NE and SE Trade winds around the equatorial areas.

Examiner's comments:

- No need to provide details explanation of the formation of the Trade winds; Trade wind convergence is used to account for the westerly track of tropical cyclones.
- Need to address the temporal distribution with attention paid to the difference in timing of tropical cyclones in the northern vs the southern hemisphere – which coincides with the summertime of the respective hemispheres.

(c) Explain how climate and vegetation lead to differences in storm hydrographs of the humid and arid tropics. [6]

Award a maximum of 3 marks for the explanation of the subdued storm hydrograph of the humid tropics.
Award a maximum of 3 marks for the explanation of the flashy storm hydrograph of the arid tropics.

Possible response:

Humid Tropics
In the humid tropics (e.g., tropical rainforests), high and consistent rainfall leads to a hydrograph with a **long lag time** and a **gentle rising limb**. This is a result of the multiple hydrological roles of dense vegetation.

- **Interception:** The multi-layered canopy intercepts a significant amount of rainfall before it even reaches the ground. This reduces the total amount of water reaching the ground surface. Interception also prevents rainsplash leading to inwashing of fines and the formation of a sealed crusted surface which can significantly reduce infiltration. Full impact of falling raindrops are absorbed by intercepting vegetation preventing compaction.
- **Soil Permeability:** The thick vegetation cover and high organic content lead to the development of deep, well-structured soils with high permeability. The roots of trees and plants create channels and pores in the soil, allowing water to infiltrate quickly and move as throughflow.
- **Antecedent Moisture:** Antecedent moisture while high, is often taken away by vegetation through photosynthesis and transpiration; thus soil moisture storage capacity is renewed quickly, allowing for infiltration in subsequent storms.
- **Proportion of Flow:** In the humid tropics, the majority of storm water moves through **infiltration** and **baseflow**. The proportion of overland flow is very low, which mitigates flood risk and leads to a gentle, "subdued" hydrograph.

Arid Tropics
In the arid tropics, the climate is characterized by low, infrequent, and often high-intensity rainfall. This creates a "flashy" hydrograph with a **short lag time** and a **steep rising limb**.

- **Interception:** The sparse vegetation provides minimal interception. The majority of rainfall hits the ground directly with high impact, which can compact the soil surface and further reduce permeability. Here inwashing of fines creates a sealed crusted surface.

- **Soil Permeability:** The lack of vegetation cover, low organic content, and baked, dry soils result in very low permeability. The ground can be hard and sealed, making it difficult for water to infiltrate. The soil cannot absorb the water fast enough, and the lack of vegetation means there's little to slow the flow.
- **Proportion of Flow:** In the arid tropics, the dominant process is **overland flow**. The high intensity of rainfall, combined with the low infiltration capacity of the soil, means that most of the water flows quickly over the surface and into the river channel. This high proportion of rapid surface runoff results in a sudden, sharp increase in river discharge, which is characteristic of a "flashy" hydrograph and increases the risk of flash flooding.

Examiner's comments:

- To be alert to the fact that storm hydrograph control factors are not the same as those affecting annual hydrographs.
- Rainfall total and rainfall intensity are two separate thing. For the storm hydrograph, focus is on intensity, while it is more relevant to focus on rainfall total either annual or monthly for the annual hydrograph.

(d) Compare the fluvial features of the meandering stream along the cross-section labelled P – Q in Resource 7 with those along the cross-section X – Y of the braided stream shown in Resource 8. [4]

Award 1 mark for each of the fluvial features.
Award 1 additional mark for relevant elaboration of fluvial features.

A **meandering stream** (cross-section P-Q) has
- a single, deep, asymmetrical channel.
- The outer bank (P) is an area of erosion, characterized by a steep or undercut **cut bank** where high velocity flow scours the bank; often termed as **river cliff**
- The inner bank or the convex bank (Q) is an area of deposition, featuring a gently sloping, shallow **point bar** made of deposited sediment. A **point bar** is a crescent-shaped depositional landform, and several points bars merged forms the gentle slipoff slope on the convex bank.
- The deepest part of the channel, the thalweg, is located near the cut bank on the outside of the bend. ~~This is due to the process of **helicoidal flow**~~.

In contrast, a **braided stream** (cross-section X-Y) has
- a wide, shallow, and asymmetrical channel that is divided into multiple smaller, interconnected channels.
- These channels are separated by exposed, temporary deposits of sediment called **mid-channel bars** or **eyots;** composed of a mix of sediments, often coarse gravel and sand, which can be poorly sorted.

Examiner's comments:

- Name and identify the fluvial landforms; not just describe the processes; in fact, description of the fluvial processes is not relevant for this question.
- Vegetation – is not a fluvial feature.
- Comparison of width of river, the depth of the river and river velocity is not relevant as the Resource do not have any reference or measures that allow for these features to be deduced.

(e) With reference to Resource 8, explain the factors that contribute to the formation of the braiding pattern shown. [6]

High Sediment Load: Braided rivers are characterized by a very high sediment load, particularly of coarse material like gravel and sand. The sediment supply often exceeds the river's capacity to transport it in a single channel, leading to deposition. In the case of Resource 8, the river is located on a glacial outwash plain, which is a key source of a large and constant supply of sediment from the melting glacier.

Variable Discharge: The flow of water in braided rivers is highly variable, with significant fluctuations between high and low flow periods. During high discharge events (egs. from seasonal melting or the snow-capped mountains in the background of Resource 8 or from monsoon rainfall during the wet season), the river has enough energy to transport a large volume of sediment and erode its banks. As the discharge decreases, the river loses energy, causing the transported sediment to be deposited. This cycle of erosion and deposition is crucial for the formation of the bars that separate the channels.

Steep Gradient: Braided streams typically form in areas with a relatively steep gradient, which gives the river a high energy level and allows it to transport the coarse sediment. This is a common feature in mountainous or glacial regions as the river upon exiting from the mountains encounters much lower river gradient and deposits its load.

A second aspect of gradient is the steep surrounding valley slopes. Such steep slopes will have low shear strength, and any slope failure will add an excessive volume of sediment load to the river at the base beyond the competence and transporting capacity of the river leading to deposition.

Erodible Banks: The riverbanks in these environments are often composed of easily erodible, non-cohesive sediment (like sand and gravel). This makes it easy for the river to widen and shift its course, which is essential for the lateral migration and frequent channel changes characteristic of a braided pattern. The sparse vegetation on the banks also contributes to their instability.

Examiner's comments

- As the focus of the question is on FACTORs – avoid excessive details on the processes leading to the formation of the braided river and its features like the mid-channel bars or eyots.
- Just stating that the low gradient is the main cause of braiding is inaccurate. It is a change from a steep to a gentle gradient (eg. the case of the Brahmaputra around the Assam area) that leads to deposition and, hence braiding.
- While variations in rainfall over different times of the year can lead to braiding, Resource 8 shows a snow-capped mountain and the title stated that the river lies on a glacial-outwash plain (meaning the river is getting it discharge from glacier meltwaters). High discharge occurs during spring and summer, while the reduce melting in winter lowers the discharge at that time.

(f) Describe four ecosystem services provided by deltas in the humid tropics.

[4]

Award one mark for each identified and explained ecosystem services provided by deltas.

Possible response:

Deltas in the humid tropics provide a variety of essential ecosystem services, which can be categorized into four main types:

1. Provisioning Services

- These services involve the production of food and fresh water. Deltas are extremely fertile due to the deposition of nutrient-rich alluvial sediment from rivers. This supports highly productive agriculture, particularly for staple crops like rice.
- The delta's network of channels and brackish water also acts as a critical habitat for a wide variety of fish, shellfish, and other aquatic life, providing a significant source of food for both local and global markets.
- Raw Materials: This includes timber for construction, cotton for textiles, and natural fibers.
- **Medicinal Resources**: Many plants and animals in ecosystems are the source of traditional and modern medicines. For instance, the rosy periwinkle from Madagascar provides compounds used to treat certain cancers.

2. Regulating Services

Regulating services are the benefits obtained from the natural regulation of ecosystem processes.

- Deltas, especially those with extensive mangrove forests and wetlands, act as natural barriers that regulate coastal environments. They stabilize shorelines and protect against coastal erosion, storm surges, and tsunamis by absorbing wave energy.
- Air and Water Purification: Wetlands and vegetation filter pollutants from water and air, improving quality.
- Climate Regulation: Forests and oceans absorb and store carbon dioxide, helping to mitigate climate change.

3. Cultural Services

- These are the non-material benefits people obtain from ecosystems through spiritual enrichment, recreation, and aesthetic experiences.
- The unique landscapes and rich biodiversity of deltas offer opportunities for ecotourism, research, and education.
- Local communities often have deep cultural and historical ties to the delta, using it for fishing, traditional practices, and spiritual ceremonies.

4. Supporting Services

Supporting services are those necessary for the production of all other ecosystem services. These fundamental processes maintain the fertility of the delta, ensuring the long-term sustainability of the provisioning and regulating services it provides.

- Nutrient Cycling: The decomposition of organic matter by microbes and fungi returns nutrients like nitrogen and phosphorus to the soil, which is necessary for plant growth.

- Soil Formation: The long-term process of breaking down rock and organic material creates the fertile soil needed for agriculture.
- Photosynthesis: This process, carried out by plants, produces the oxygen we breathe and is the basis for all food webs.
- Habitat Provision: Ecosystems provide a home for all species, which in turn supports biodiversity and the complex web of life that makes other services possible.

Section B

Cluster 1 Development, Economy and Environment

3 Evaluate the strategies to manage conflicts over transboundary water resources. [20]

Suggested Approach:
Candidates can approach the question by evaluating whether strategies such as international treaties, cooperative river basin commissions, economic cooperation, and diplomatic mediation have effectively managed tensions over shared water resources. This can be assessed through case studies such as the Colorado River between the USA and Mexico, the Mekong River across Southeast Asia, the Jordan River in the Middle East, and the Okavango River in Southern Africa. Candidates should analyse whether these strategies have achieved their intended aims of preventing open conflict, maintaining water quality, and ensuring equitable distribution of water while balancing development and environmental needs. They could also compare the relative effectiveness of bilateral versus multilateral agreements and assess how institutional capacity, power asymmetries, and political will shape the outcomes of these strategies. The evaluation should consider both the successes and persistent challenges, especially in light of climate change and competing developmental priorities.

Indicative Content:

One important strategy has been the negotiation of international treaties and agreements. The 1944 Colorado River Treaty between the United States and Mexico demonstrates how bilateral treaties can provide clarity and stability. The treaty allocated specific volumes of water to Mexico and established the International Boundary and Water Commission (IBWC) to manage disputes. The durability of this arrangement, even when water quality deteriorated in the 1960s due to salinity problems, highlights its effectiveness. When Mexico formally protested, the United States renegotiated terms through the IBWC in the 1970s, guaranteeing water quality improvements. The success of this treaty is partly explained by the limited number of riparian states, the relatively cooperative relationship between the two countries, and the existence of a strong institution with the capacity to resolve disputes.

A second strategy has been the creation of cooperative river basin commissions. The Permanent Okavango River Basin Water Commission (OKACOM), established in 1994 by Angola, Namibia, and Botswana, illustrates how institutionalised cooperation can help prevent escalation. OKACOM was tasked with promoting equitable allocation and sustainable use, commissioning joint environmental assessments, and providing a forum for dialogue before conflict could intensify. The commission has helped defuse tensions, such as when Namibia planned a pipeline to divert water to Windhoek during the 1990s drought, by fostering dialogue and mobilising international support. However,

its effectiveness is limited by the lack of enforcement mechanisms, reliance on consensus, and the option for states to withdraw.

Economic cooperation and dominance also play a role in transboundary water management. China's position as the upstream riparian in the Mekong has enabled it to pursue unilateral dam building, while simultaneously using aid, trade, and infrastructure investment to maintain influence over downstream neighbours. This strategy reduces the likelihood of violent conflict but entrenches asymmetrical power relations, raising questions about equity and long-term sustainability. At times, economic cooperation frameworks such as the Greater Mekong Subregion programme have fostered connectivity, energy sharing, and poverty reduction, suggesting that shared economic benefits can encourage collaboration. However, such cooperation often sidelines environmental concerns, and during crises such as the 2010 Mekong drought, diplomatic negotiation rather than economic incentives became the more effective strategy to manage tensions.

In evaluation, the effectiveness of strategies to manage transboundary water conflicts depends heavily on institutional capacity, political commitment, and the relative power of riparian states. Where upstream powers show willingness to compromise, treaties and commissions can achieve long term stability. However, where institutions are weak, or where broader political or strategic hostilities dominate, agreements struggle to succeed, as seen in the Jordan Basin and the Mekong. Economic cooperation strategies may provide short-term stability but risk perpetuating inequities, while negotiation and mediation are only as effective as the political will behind them. Overall, strategies have largely succeeded in preventing violent conflict, but their success in achieving equitable and sustainable water management remains mixed.

4 Evaluate the relative influence of the multilateral institutions in shaping the global economy. [20]

Suggested Approach:
Candidates can approach the question by evaluating whether multilateral institutions such as the World Bank, WTO, and ASEAN continue to exert decisive influence over the global economy, or whether their influence has become more limited compared to the role of powerful states. Candidates should analyse the ways in which these institutions shape global finance, trade, and regional integration, and then weigh this against the ability of individual states to set agendas, implement industrial policies, and negotiate economic arrangements. A strong answer would explore the effectiveness and limitations of these multilateral institutions, highlight how their influence varies across different contexts, and conclude with a reasoned judgement about whether the global economy is primarily shaped by multilateralism or by the actions of states.

Indicative Content:
The World Bank has long shaped development through its lending policies, technical advice, and project financing. Its role in providing loans for infrastructure, education, and health in the Global South has allowed it to influence the direction of economic growth. Conditionalities tied to its loans often push recipient countries toward structural reforms and governance improvements. For instance, structural adjustment programmes in the 1980s and 1990s tied access to credit to liberalisation and privatisation, embedding neoliberal economic models across much of the developing world. However, the Bank's influence is contested. Critics argue it entrenches dependency and reflects the interests of major donor states, particularly the US, whose voting power is disproportionate. More recently, the rise of alternatives such as China's Belt and Road Initiative has weakened the Bank's monopoly, showing its influence is no longer uncontested.

The World Trade Organization (WTO) plays a central role in regulating the rules of trade and promoting liberalisation. Its dispute settlement mechanism has been critical in managing trade conflicts and enforcing compliance, giving it authority to shape how countries engage in the global economy. For example, the WTO has arbitrated disputes involving intellectual property, subsidies, and tariff barriers, influencing the policies of both developed and developing nations. Yet, the WTO's influence has declined in the twenty-first century. The Doha Development Round has stalled, and member states increasingly pursue bilateral or regional trade agreements, bypassing multilateral negotiations. Rising protectionism, such as US-China trade tensions, has also undermined the WTO's ability to uphold liberalisation. This demonstrates that while the WTO retains formal authority, its practical influence has diminished relative to the power of states to pursue independent trade strategies.

Regional organisations such as ASEAN also play a role in shaping the global economy by fostering regional integration and collective bargaining power. Initiatives like the ASEAN Free Trade Area (AFTA) and ASEAN Economic Community (AEC) aim to reduce barriers and harmonise rules across Southeast Asia, increasing competitiveness and attracting foreign investment. ASEAN has also been central in negotiating mega-regional agreements such as the Regional Comprehensive Economic Partnership (RCEP), which includes China, Japan, and Australia alongside ASEAN members, creating the world's largest trading bloc by population. However, ASEAN's consensus-based decision-making and diverse economic structures among members often limit its ability to act decisively. Its influence is thus strongest at the regional scale but remains secondary when compared to global actors such as the WTO.

When compared to these institutions, the role of the state remains highly significant in shaping the global economy. Powerful economies like the United States, China, and the European Union often set global norms and exert leverage beyond multilateral frameworks. Similarly, the US has at times bypassed multilateral institutions, using its economic weight to impose sanctions, negotiate bilateral deals, or shape financial markets. Even within multilateral organisations, states retain ultimate authority, as voting rights and decision making power in the national scale. This demonstrates that while multilateral institutions provide frameworks and rules, states continue to exercise decisive influence when national interests are at stake.

In evaluation, multilateral institutions such as the World Bank, WTO, and ASEAN play important roles in setting norms, providing finance, and facilitating cooperation, but their influence is often constrained by the power of states. Institutions provide platforms for negotiation and rules-based order, yet they depend on the willingness of states to comply. In practice, the global economy reflects a dynamic interplay: institutions provide structure and predictability, but states often act unilaterally or bilaterally to secure their interests. The relative influence of multilateral institutions, therefore, remains significant but increasingly contested, and their ability to shape the global economy cannot be understood in isolation from the power and agency of states.

Cluster 2 Tropical Environments

5 'Fluvial flooding in the humid tropics is largely the result of human activities in the drainage basin.' Evaluate this statement. [20]

Possible answer

The argument can be framed based on this one suggested conclusion:

Fluvial flooding in the humid tropics cannot be attributed solely to human activity. Climatic factors such as monsoon rainfall, tropical cyclones, and La Niña anomalies provide a powerful natural basis for flooding. However, human modifications of drainage basins—deforestation, rapid urbanisation, and mismanaged flood defences—commonly amplify flood magnitude and damage. The statement that flooding is "largely the result of human activities" is *partially true*: in densely urbanised areas like Jakarta, Chennai and Singapore, human factors dominate, while in naturally flood-prone basins like Bangladesh, natural causes remain more significant. Ultimately, it is the *interaction* between natural and human processes that explains why floods in the humid tropics are increasingly severe. It could also be argued that natural factors are the main cause of flooding in the humid tropics, since excessive water almost always comes from the input of precipitation. Human activities exacerbate the flooding and are not the cause of fluvial flooding in the humid tropics.

Human Activities that Exacerbate Flooding
While natural drivers set the stage for flooding, human activity in drainage basins often worsens both magnitude and impacts.

1. **Deforestation and Land Use Change**
   Deforestation reduces infiltration and interception, while increasing surface runoff. In **Thailand (2011)**, deforestation in upstream basins contributed to faster runoff into the Chao Phraya, overwhelming downstream defences. Similarly, in **Malaysia (2014)**, logging in Kelantan's headwaters reduced forest cover and increased sedimentation, decreasing river capacity. Flooding in Bangladesh and Pakistan drainage basins are also attributed deforestion in the upper basin particularly in the Himalayan region. Talibans to fund their war efforts, carried out illegal logging in the upper basins of the Indus River for example.

2. **Urbanisation and Encroachment**
   Rapid urbanisation creates impermeable surfaces and restricts natural drainage. In **Chennai, India (2015)**, severe floods were not only due to monsoon rainfall but also extensive urban development on natural floodplains and wetlands. In Chennai, in push for more urban land, many wetlands and surface stores like lakes are filled up. This caused the loss of natural stores in Chennai, and reduce the natural storage capacity of the drainage basins of the Adyar and Cooum Rivers. Storm drains and channels were poorly maintained, while lakes that once acted as natural sponges had been reclaimed for construction. Likewise, in **Singapore (2010)**, heavy rainfall (100 mm in two hours) caused Orchard Road to flood because drains were undersized and blocked by urban development. This highlights how even technologically advanced cities in the humid tropics are vulnerable when natural drainage is compromised.

3. **River Management and Infrastructure**
   - Poorly managed flood defences can backfire. In **Bangladesh (1998)**, embankments and polders confined rivers, raising water levels and making breaches catastrophic.

   - In **Thailand (2011)**, upstream dam releases worsened downstream inundation when operators were forced to release excess water suddenly. These examples show that structural measures can sometimes intensify rather than reduce flooding. While human activities exacerbate the flooding, the 2011 flood of Thailand arose from the fact that nature had released large amount of rainfall into the Chao Praya basin for an extended period of time prior to the flooding. There were five consecutive tropical storms in 2011 that contributed to the heavy rainfall and floods in Thailand. These storms, which occurred between the end of June and the beginning of October, included Nock-Ten and caused heavy precipitation in the northern and central regions of the country.

   - Another example of large volume of water released from dams include the **Brisbane flood of 2011** where water from the Wivenhoe Dam was released to avoid the breach of the dam from the intense rainfall. This is an excellent example of how nature through the high rainfall and long duration rainfall that started at the end of 2010 induced by La Nina, lead to excessive input of water into Brisbane Basin in Queensland. The opening of the gates the Wivenhoe dam increased the severity of the flooding in the drainage basin. The Lockyer and Bremer Rivers were not able to discharge their waters into the main trunk stream of the Brisbane River leading to the backup of water within these two tributaries of the Brisbane River, causing flooding in the basins of these two tributaries.

   - In the **north Pakistan flood of Aug-Sept 2025**, the intense summer monsoon rainfall once again is responsible for the high input of rainfall into the northern Indus Basin. Additional water was added to the flood due to barrages of the Indus River in the Indian territory breaking and releasing additional water downstream. Pakistan federal minister accused New Delhi (capital of India) of deliberately releasing excess water from their upstream dams without timely warning. Yet it should be noted that when there is high input into any dams or barrage from melting glaciers, intense summer monsoon water has to be released from such structures or the entire structure is at risk of complete failure. The barrages could also have failed due to these dams being built for a much smaller capacity and now currently cannot cope with the high rainfall arising from climate change. This is yet another example of natural factors operating and human activities making the flood more severe.

4. Subsidence

Use example of subsidence in Jakarta and Bangkok which arise from the excessive withdrawal of groundwater leading loss of pore pressure and hence subsidence. These lower locations within Jakarta where the Ciliwung River flows through becomes the collection point for any excessive water flowing through. When high rainfall input coincides with high tide, both coastal and fluvial flooding occur in Jakarta.

And alternative approach
Use a few large examples of floods (2 to 3 examples) and for each explain the contribution of natural and human factors.

Examiner's comments

- Not necessary to include long explanations of how tropical rainfall is formed through Hadley cell, monsoon, tropical cyclones and ENSOs.
- Avoid just name-dropping – just country names will not do; eg. India, Indonesia
- Indus River is in largely in Pakistan, although some northern parts of it is in India.
- Ganges and Brahmaputra is largely in Bangladesh in its lower course, and many other countries in its upper course.
- Ensure accuracy of flood events, and flood details. Gross errors will mean the given example will not be credited.

6 Evaluate the factors influencing the formation of karst landforms in the humid tropics. [20]

Suggested approach

Candidates should explain the role of various factors including climate, geological, vegetation, tectonic and climate change which usually operates in tandem to create tropical karst landforms in locations like, Bohol of the Phillippines, Cockpit Country in Jamaica, Krabi of Thailand, Halong Bay of Vietnam, and the cone and tower karst of Guangxi in south China. Explanations involving processes of differential carbonation, tectonic uplift, lateral planation by river and waves, basal carbonation and mass movement should be incorporated to show how the range of factors affect geomorphic processes to create cone and tower karsts and the caves associated with them.

Possible response

Key karst landforms in the humid tropics include cone and tower karsts and caves. Cone or cockpit karsts consist of star-shaped depressions called cockpits separated from each other by roughly equally spaced conical/cone-shaped hills (cone karst) with local relief that is anything from 30 to 300 metres. The landscape has often been labelled as one with an egg-box topography. The cones or conical hills are generally symmetrical, and their slopes measured between 45° and 60°, regardless of their structure. Conical hills are 30 to 120 m tall and occur with a density of 15 to 30 hills/km^2. Example of cockpit karst is Chocolate Hills, Bohol, Philippine, and the cone karst in Guilin and Yangshuo, Guangxi, South China. Those in Guangxi for example are 30 - 80 metres tall and have slope angles of 45–55°.

The drainage of cone karst is almost entirely underground, entering either by fissure percolation or through open sinks at the ends of short ephemeral stream courses within the dolines. Water tables commonly lie far beneath (more than 100- 300 metres deep) the doline floors is, and long vadose cave passages can form dendritic systems. In the case of tower karst in south China, the water table is now stable and around the height of the alluvial karst plains.

Tower karst are hills that are steeper and higher than cone karst and have a bigger height-diameter ratio. These residual hills come in a range of shapes, though a common dominant feature is their sharp rise from the surrounding plains. Their heights may reach over 150 m in Sarawak and even 300 m elsewhere such as Vietnam and Guangxi, China. They occur with a density of 5-10 hills/km^2 . Tower karst in Guangxi are as tall as 300 metres with individual slopes ranging from 60° to 75°. Similarly, the tower karst in Thailand, like those located at Krabi, are narrowed base with steep flanking slope

reaching up to 200 metres in height. Erosional notches are found at the base of many tower karst as those in South China and those in coastal Thailand and Vietnam (tower karst standing in seawater in Halong Bay).

In locations with cone and tower karst, caves are another typical landform of the karst region. Caves in the humid tropics tend to be smaller although there are exceptionally large ones in the Mulu region, like the Sarawak Chamber, the Deer Cave and the Clearwater Cave linked to an underground river. Similar large caves are also found in the karst region of Indochina: Phong Nha-Ke Bang National Park and the Hin Nam No National Park boasting the world's oldest and largest karst systems.

Karst landforms are the product of chemical weathering of carbonation, river and marine erosion, biological activities, and tectonic activities combined together to give rise to a range of cone and tower karsts and their caves. Factors that affect these geomorphic processes include climate, geology, biologic and tectonic factors that work synergistically to create the karst landforms in the humid tropics.

The geological factor plays a large role in karst landform formation. Karst landforms commonly develop in carbonate rocks like limestone and dolomite with high purity and with sufficient thickness. The thickness of the limestone allows for high relief of conical hills and tower karst to form as carbonation deepens the intervening cockpits. A huge thickness of limestone, sufficient to allow massive surface lowering, without reaching the base of the limestone, which gives time and space for the karst landscapes to evolve to maturity as in the case of the tower karst in Yangshuo and Guilin, Guangxi, south China.

The hummocky or egg-box topography of cone or cockpit karst arise from differential rates of carbonation. Limestone has the property of low primary permeability since it is crystalline rock. Instead, water enters the rock through joints and bedding planes. Due to the lack of primary permeability, limestone carbonation is focused along joints and bedding planes which provide secondary permeability. Especially at the intersection of joints and/or bedding planes, accelerated carbonation enlarge these gaps, creating dolines which over time enlarge to become star-shaped cockpits. Deepening of the cockpits continues until the water table is reached. The high concentration of dissolved calcium carbonate in groundwater reduces the rates of carbonation.

The limestone surrounding the cockpits undergo less erosion as water runs down the conical hills rapidly giving little time for extended carbonation. The less weathered parts are left and the conical hills. Water running down the hillslope tend to take preferred paths, forming gullies. These gullies aid in the formation of the star-shaped cockpits. The water running down the conical hills collect in the cockpits leading to it enlargement and deepening. The neat alignment of hills and corridors in the Chocolate Hills of Bohol and those in Cockpit Country in Jamaica are attributed to the presence of intersecting joints and faults. This highlights the importance of the geological factor in determining the formation of cone karst and their alignment.

A second way in which geology guides formation of tropical karst is the purity of the limestone and this is best demonstrated by the cockpit karst of Jamaica. Here, the White Limestone leads to the development of cockpit karst with conical hills, but not the less pure Yellow Limestone which leads to doline formation only.

While geological control is strong, it is also important for the right climate condition to exist. The humid tropics provide the perfect climatic conditions for accelerated karstification. The first and most critical factor is the abundant precipitation, which acts as the primary agent of dissolution. Rainwater absorbs atmospheric carbon dioxide to

form a weak carbonic acid, which then reacts with the calcium carbonate in limestone dissolving into calcium bicarbonate which in turn is removed in solution. The dissolved parts often involve joints and bedding planes which are enlarged by the process of carbonation.

The second crucial climatic factor is the consistently high temperature. Chemical reaction rates are directly proportional to temperature, a principle encapsulated by the Van't Hoff rule. This means that the tropical climate, with its average temperatures of over 25°C, significantly speeds up the dissolution process. The continuous supply of warm, acidic water ensures that chemical weathering is a year-round, highly efficient process. Thus, cone and tower karst only form in locations like the humid tropics. The importance of high rainfall and temperature is particularly highlighted by the fact that locations with hot desert climate like northern Africa (Sahara Desert) are underlain by carbonate rocks, but cone karst and tower karst do not form there due to the lack of water for the carbonation process.

The humid tropics also provide conditions that enhance the carbonation process. The climatic conditions in the humid tropics promote a dense forest cover. Biological activity enhances the acidity of the water. The dense rainforest vegetation and rich, organic soils of the humid tropics are key contributors. As plant matter decays, it releases **humic acids**, which are more potent than carbonic acid in dissolving limestone. Additionally, root respiration and the decomposition of organic material led to a very high partial pressure of CO_2 in the soil air. This concentration can be up to 100 times greater than in the atmosphere, creating a more aggressive carbonic acid solution as water percolates through the soil. For the tower karst in Thailand, the deep cockpits are suggested to be the result of intense carbonation occurring at localised spots likely at the intersection of vertical joints and/or bedding planes.

For some humid tropic locations, tectonic activity and climate change have contributed to the formation of tower karst. In Guangxi, south China, the tower karst are regarded as an evolution of the cone karst in the same area. The slow tectonic uplift of the region allows for the water table to be lowered. Slow tectonic uplift that matches surface denudation allowed the karst plain to be lowered through the limestone layers. As the depth of carbonation slows dramatically at the water table since groundwater there is saturated with dissolved calcium bicarbonbate, cockpit deepening would be halted. The tectonic uplift led to a lowering of the water table allowing for the continued deepening of the cockpits, while surrounding conical hills ‘grow’ taller to become tower hills in Guilin and Yangshuo in South China. When the new lowered water table is reached, this forms the new base level of erosion, vertical deepening at the cockpit slows down. Instead, lateral planation take over.

Lateral planation occurs largely at the water that flows down the hills collects at the base where the cockpits are covered with a thin layer of alluvium. This freshwater thus attacks the base of the hills through carbonation creating an overhang or basal erosional notch. In Guilin and Yangshuo, lateral erosion by the Li River also contributes to basal undercutting. Limestone hills with vertical joints or vertical bedding planes when there is excessive removal of toe support with the formation of the erosional notch leads mass movement, often in the form of rockfall or rockslide facilitated by the lines of weakness of joints or bedding planes. As lateral planation proceeded, the intervening hills are reduced in size and some are completed eroded away, while the cockpits merged to form an extensive alluvial plain. In the case of tower karst in Guangxi, China, the combination of the tectonic factor and the geological lines of weaknesses lead to the formation of steep-sided hills separated by a vast alluvial plane, ie, tower karst landscape.

While lateral planation may have played a similar role in the formation of the tower karst in Krabi and in Halong Bay, an additional geomorphic processes currently continues to shape the tower karst. The climatic factor comes into play in the form of climate change. Low-lying coastal locations in Indochina and Southeast Asia has experienced around 120 metres of sea level rise since the onset of the interglacial after the last ice age around 18,000 years ago. The alluvial plains surrounding the tower karst around Krabi and Halong Bay were submerged by this episode of sea level rise, leaving the tower hills as small islands standing in the sea. Wave erosion takes over as the current erosional process attacking the base of the cliffs of tower karst creating the erosional notch, and continuing the erosion of coastal caves. Excessive overhanging cliffs with large erosional notches periodically undergo rockslide and rockfall along inherent lines of weakness like joints and bedding planes, reducing the diameter of the tower hills, and at time complete elimination of the hills.

Tectonic uplift also accounts for the dry caves in tower karst. Caves form best when in constant contact with freshwater. At the surface of water table, there is more freshwater compared to greater depth. Carbonation along joints and bedding planes enlarge the cavity, and further rockfall from the roof the caves can further enlarge the caves. Often caves are also passage of subterranean rivers, and fluvial erosion also contributes to cave formation. Karst at coastal locations also have wave erosion as additional geomorphic process that form and enlarge caves. Speleothems like stalactites and stalagmites form in caves when the water carrying dissolved calcium carbonate re-precipitate back into a solid in the cave atmosphere where carbon dioxide readily escape the water droplets hanging on the roof of caves or those the splash onto the cave floors.

Climate and geology are two critical factors in the formation of tropical karst of cone and tower karst. While it is possible to form isolated hills with carbonate rocks, the distinctive cone and tower karst landscape are typically associated with carbonate rocks like limestone. The thickness and the purity of limestone in location like Jamaica are another added Climate plays an equally important role; in locations where there is extensive carbonate rocks like in northern Africa, the low rainfall conditions prevent the formation of tropical karst. The dense tropical forest arise also from the high rainfall and temperature of the humid tropics and this adds to the intense carbonation that leads to cone and tower karst formation in the case of Thailand. In selected locations like Guangxi, the tectonic process plays a critical role in transforming cone karst into tower karst. In Krabi and in Halong Bay, changes in sea level rise due to a change to an interglacial climate continues to shape tower karst standing in the sea. The conclusion would be that a range of factors, together with earth forces act together to create the unique humid tropical karst landforms.